

Gluco Twin 5-Layer End-to-End Architecture

LAYER 01

Data Ingestion

EHR, 15-minute wearables, CGM streams, and nutrition records.

LAYER 02

Feature Engine

Rolling statistics, trend vectors, meal lags, and metabolic interactions.

LAYER 03

Digital Twin Core

State history, recalibration engine, sleep/stress context, and state cloning.

LAYER 04

ML + XAI

XGBoost spike risk probability and SHAP TreeExplainer attribution.

LAYER 05

Clinician UI

FastAPI backend, React dashboard, signal charts, and what-if simulation sliders.